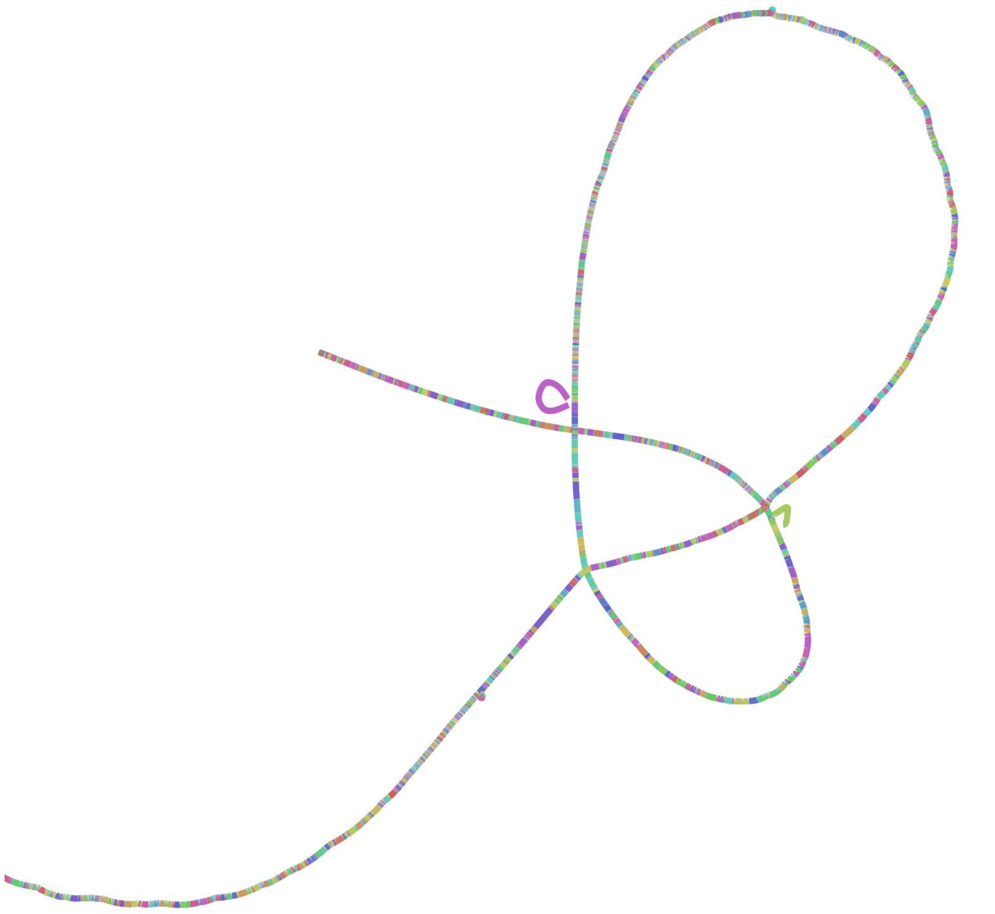
Minigraph Cactus

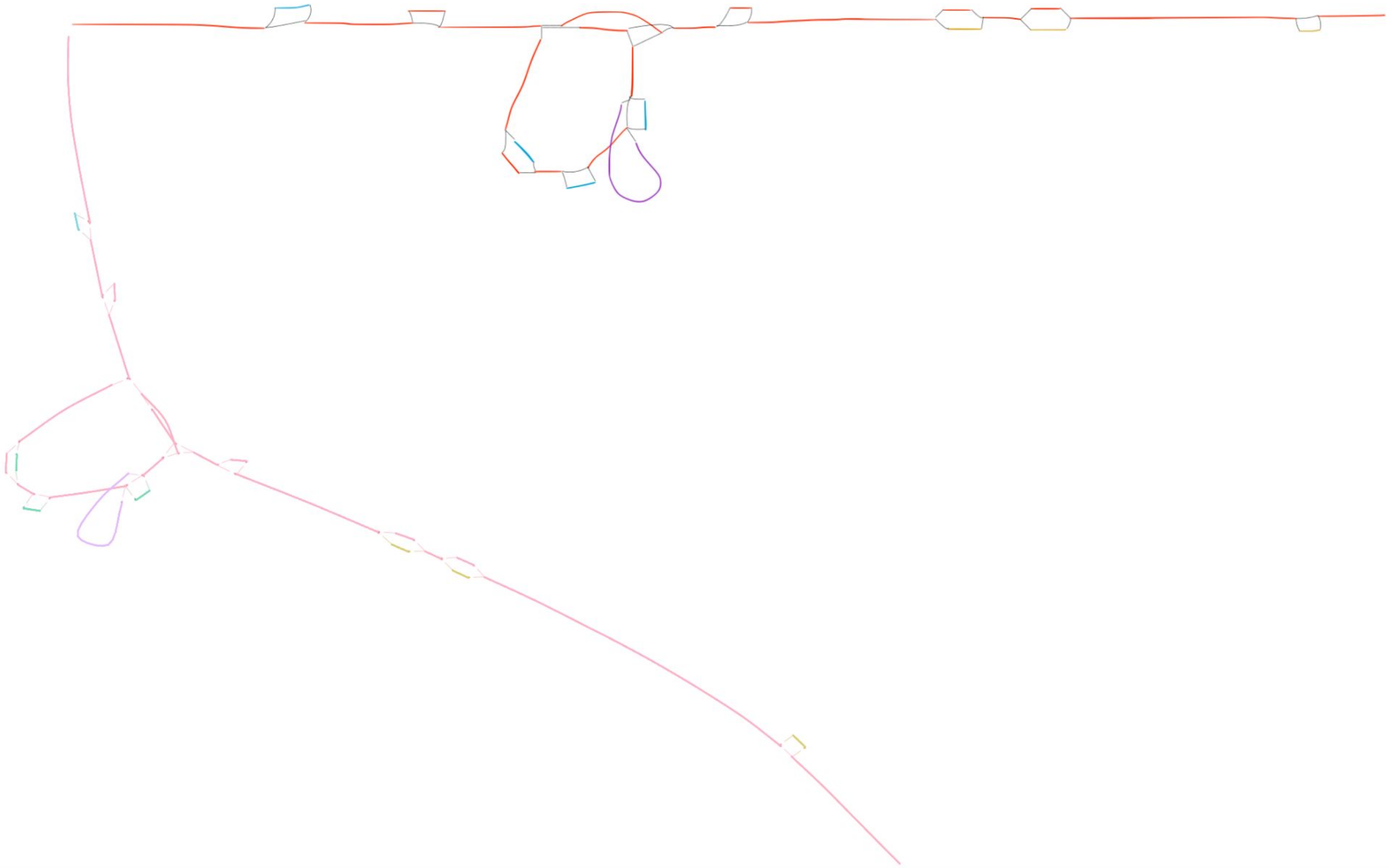
- graphtype: “MC”
- debug_small_graphs: false
- minnodelen: 1.0
- nodeseglen: 1.0
- edgelen: 0.1
- nodelenpermb: 1000000

The default setting for most of these variables may make the minigraph cactus graph look really bad, so we may need to do some experiments on that. (the setting above can be a start point)



Bandage Variables

- graphtype: Minigraph (the one we have implemented now), or Minigraph Cactus
- debug_small_graphs: (bool) if true, each node's drawn length is exactly the node's length in base pair
- minnodelen: (float) minimum node length to draw. For every node, if it's drawn length is smaller than **minnodelen**, we use **minnodelen** as their drawn length. <u>(default: 5.0, min: 1.0, max: 100.0)</u>
- nodeseglen: (float) max length of every OGDF nodes. (eg. if node length is larger than **nodeseglen**, the node will be splitted into several OGDF nodes). <u>(default: 20.0, min: 1.0, max: 1000.0)</u>
- edgelen: (float) edge length between nodes <u>(default: 5.0, min: 0.1, max: 100.0)</u>
- nodelenpermb: (float) node length per megabase. A scaling factor for the node length drawn. If **debug_small_graph** is false, we use **nodelenpermb** to calculator the node length that will be drawn. <u>(default: 1000.0, min: 0, max: 1000000.0)</u>
    - drawnNodeLength = NODELENPERMB * Node Length in bp /1000000
    - Setting **nodelenpermb** to 1000000 has the exact same effect as setting **Debug_small_graphs** to true (it's better to use nodelenpermb, and I may delete debug_small_graphs eventually)



Sept. 19th, 2025

Timeline

- **Metadata**
    - Color by **<u>superpopulation</u>** and **<u>population</u>** count
    - Light up nodes by superpopulation
        i. Color by frequencies of this superpopulation
    - Minor Adjustments:
        i. Metadata page: show assembly numbers instead of sample numbers
        ii. Assembly page: record haplotype numbers in assemblies
- **Tracks (GRCh38 track)**
- **Gene Annotation**
- **Aligning the tracks linearly**

Feb. 26th, 2026

New Data - coordinates of each node and each assembly


```
"assembly": {
    "CHM13#0": {
        "sequence_id": "chr1",
        "region": "25037160-25419794"
    },
    "GRCh38#0": {
        "sequence_id": "chr1",
        "region": "25200905-25582458"
    },
    "HG00097#1": {
        "sequence_id": "CM094060.1",
        "region": "25632490-26015120"
    },
```



```
"node": {
   "5504+": {
        "name": "5504+",
        "length": 35895,
        "assembly": [
        {
            "assembly_name": "CHM13",
            "haplotype": "0",
            "sequence_id": "chr1",
            "path_strand": "+",
            "node_strand": ">",
            "start": "25037160",
            "end": "25073056"
        },
```

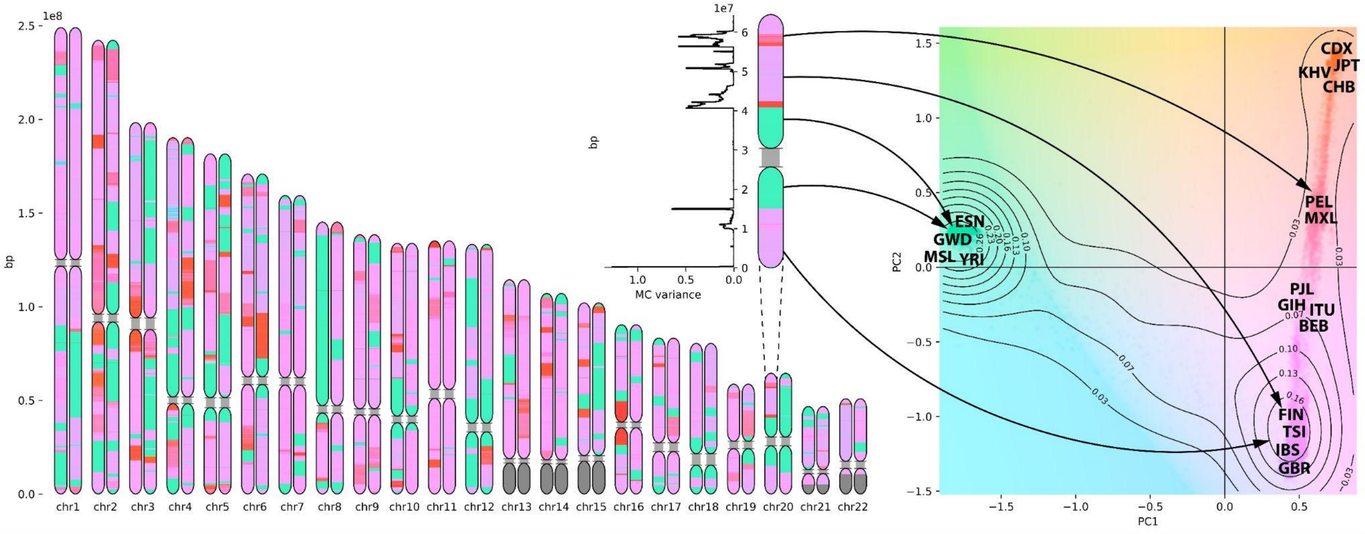

- Is this information enough for rendering the gene annotation interactively?
- Show the queried assembly region somewhere on the screen

What is the best way to pass the gene annotation data of each assembly?

```
[xbu@login1 gene_annotation]$ zcat HG01081_pat_hprc_r2_v1.0.1_cat_v1.1.gff3.gz|head -n 10
##gff-version 3
HG01081#1#CM088564.1	CAT	gene	21373	24722	.	+	.	ID=HG01081_pat_G000001;gene_id=ENSG00000302559.1;gene_name=ENSG00000302559;gene_type=lncRNA;
level=2
HG01081#1#CM088564.1	CAT	transcript	21373	24722	.	+	.	ID=HG01081_pat_T000001;Parent=HG01081_pat_G000001;gene_id=ENSG00000302559.1;gene_nam
e=ENSG00000302559;gene_type=lncRNA;level=2;sequence_identity=0.995;tag=basic,Ensembl_canonical,GENCODE_Primary,TAGENE;transcript_id=ENST00000787856.1;transcript_name=ENST00
000787856;transcript_type=lncRNA
HG01081#1#CM088564.1	CAT	exon	21373	21522	.	+	.	ID=HG01081_pat_E000001;Parent=HG01081_pat_T000001;exon_id=ENSE00004165835.1;exon_number=1;ge
ne_id=ENSG00000302559.1;gene_name=ENSG00000302559;gene_type=lncRNA;level=2;tag=basic,Ensembl_canonical,GENCODE_Primary,TAGENE;transcript_id=ENST00000787856.1;transcript_nam
e=ENST00000787856;transcript_type=lncRNA
HG01081#1#CM088564.1	CAT	exon	24431	24722	.	+	.	ID=HG01081_pat_E000002;Parent=HG01081_pat_T000001;exon_id=ENSE00004165834.1;exon_number=2;ge
ne_id=ENSG00000302559.1;gene_name=ENSG00000302559;gene_type=lncRNA;level=2;tag=basic,Ensembl_canonical,GENCODE_Primary,TAGENE;transcript_id=ENST00000787856.1;transcript_nam
e=ENST00000787856;transcript_type=lncRNA
HG01081#1#CM088564.1	CAT	gene	32029	40085	.	-	.	ID=HG01081_pat_G000002;gene_id=ENSG00000184731.6;gene_name=FAM110C;gene_type=protein_coding;
havana_gene=OTTHUMG00000151321.2;hgnc_id=HGNC:33340;level=2
HG01081#1#CM088564.1	CAT	transcript	32029	39720	.	-	.	ID=HG01081_pat_T000002;Parent=HG01081_pat_G000002;ccdsid=CCDS42645.1;gene_id=ENSG000
00184731.6;gene_name=FAM110C;gene_type=protein_coding;havana_gene=OTTHUMG00000151321.2;havana_transcript=OTTHUMT00000322220.2;hgnc_id=HGNC:33340;level=2;protein_id=ENSP0000
0328347.4;sequence_identity=0.999;tag=basic,Ensembl_canonical,GENCODE_Primary,MANE_Select,appris_principal_1,CCDS;transcript_id=ENST00000327669.5;transcript_name=FAM110C-20
1;transcript_support_level=1;transcript_type=protein_coding;variation_against_ref=frameshift
HG01081#1#CM088564.1	CAT	exon	32029	34842	.	-	.	ID=HG01081_pat_E000003;Parent=HG01081_pat_T000002;ccdsid=CCDS42645.1;exon_id=ENSE00001537167
.2;exon_number=2;gene_id=ENSG00000184731.6;gene_name=FAM110C;gene_type=protein_coding;havana_gene=OTTHUMG00000151321.2;havana_transcript=OTTHUMT00000322220.2;hgnc_id=HGNC:3
3340;level=2;protein_id=ENSP00000328347.4;tag=basic,Ensembl_canonical,GENCODE_Primary,MANE_Select,appris_principal_1,CCDS;transcript_id=ENST00000327669.5;transcript_name=FA
M110C-201;transcript_support_level=1;transcript_type=protein_coding
HG01081#1#CM088564.1	CAT	exon	38655	39720	.	-	.	ID=HG01081_pat_E000004;Parent=HG01081_pat_T000002;ccdsid=CCDS42645.1;exon_id=ENSE00001292553
.4;exon_number=1;gene_id=ENSG00000184731.6;gene_name=FAM110C;gene_type=protein_coding;havana_gene=OTTHUMG00000151321.2;havana_transcript=OTTHUMT00000322220.2;hgnc_id=HGNC:3
3340;level=2;protein_id=ENSP00000328347.4;tag=basic,Ensembl_canonical,GENCODE_Primary,MANE_Select,appris_principal_1,CCDS;transcript_id=ENST00000327669.5;transcript_name=FA
M110C-201;transcript_support_level=1;transcript_type=protein_coding
HG01081#1#CM088564.1	CAT	CDS	34823	34842	.	-	2	ID=HG01081_pat_C000001;Parent=HG01081_pat_T000002;ccdsid=CCDS42645.1;exon_id=ENSE00001537167
.2;exon_number=2;gene_id=ENSG00000184731.6;gene_name=FAM110C;gene_type=protein_coding;havana_gene=OTTHUMG00000151321.2;havana_transcript=OTTHUMT00000322220.2;hgnc_id=HGNC:3
3340;level=2;protein_id=ENSP00000328347.4;tag=basic,Ensembl_canonical,GENCODE_Primary,MANE_Select,appris_principal_1,CCDS;transcript_id=ENST00000327669.5;transcript_name=FA
M110C-201;transcript_support_level=1;transcript_type=protein_coding
```

PCA Visualization

- If we need 20% of the node to be one color, and 80% of the node to be another, is it possible to do?
- PCA Chromosome/Contig view of whole chromosome/contig
- Add a track for prediction accuracy



Others

- Add chromosome/contig view and coordinates for the assemblies
- Change population visualization to one sided heatmap